

Lab 7: Page Replacement Algorithms

1. Write a program to implement optimal page replacement algorithm and calculate page faults.

```
#include <stdio.h>
int main() {
    int no_of_frames, no_of_pages, frames[10], ref_string[30], temp[10], flag1, flag2, flag3, i, j, k, pos, max, faults = 0, flag = 0;
    printf("Enter number of frames: ");
    scanf("%d", &no_of_frames);
    printf("Enter number of pages: ");
    scanf("%d", &no_of_pages);
    printf("Enter page reference string: ");
    for (i = 0; i < no_of_pages; ++i) scanf("%d", &ref_string[i]);
    for (i = 0; i < no_of_frames; ++i) frames[i] = -1;
    printf("\n=====
=====\\n");
    for (i = 0; i < no_of_frames; i++) printf("frame[%d]\\t", i);
    printf("\n=====
=====\\n");
    for (i = 0; i < no_of_pages; ++i) {
        flag1 = flag2 = 0;
        for (j = 0; j < no_of_frames; ++j) if (frames[j] == ref_string[i])
        { flag1 = flag2 = 1; flag = 1; break; }
        if (flag1 == 0) for (j = 0; j < no_of_frames; ++j) if (frames[j] ==
        -1) { faults++; frames[j] = ref_string[i]; flag2 = 1; break; }
        if (flag2 == 0) {
            flag3 = 0;
            for (j = 0; j < no_of_frames; ++j) {
                temp[j] = -1;
                for (k = i + 1; k < no_of_pages; ++k) if (frames[j] == ref_
                string[k]) { temp[j] = k; break; }
            }
            for (j = 0; j < no_of_frames; ++j) if (temp[j] == -1) { pos =
            j; flag3 = 1; break; }
            if (flag3 == 0) {
                max = temp[0]; pos = 0;
                for (j = 1; j < no_of_frames; ++j) if (temp[j] > max) { max
                = temp[j]; pos = j; }
            }
            faults++; frames[pos] = ref_string[i];
        }
        printf("\\n");
        for (j = 0; j < no_of_frames; ++j) printf("%4d\\t\\t", frames[j]);
        if (flag == 0) printf(" F");
        flag = 0;
    }
```

```

    }
    printf("\n=====
=====\\n");
    printf("\nTotal Page Faults = %d\\n", faults);
    return 0;
}

```

```

Enter number of frames: 4
Enter number of pages: 10
Enter page reference string: 0 1 7 2 3 2 7 1 0 4

```

frame[0]	frame[1]	frame[2]	frame[3]	frame[4]
0	-1	-1	-1	F
0	1	-1	-1	F
0	1	7	-1	F
0	1	7	2	F
3	1	7	2	F
3	1	7	2	
3	1	7	2	
3	1	7	2	
0	1	7	2	F
4	1	7	2	F

```

Total Page Faults = 7
PS C:\Users\samir\Desktop\project>

```

2. Write a program in C to implement FIFO page replacement algorithm and calculate page faults.

```

#include <stdio.h>
int main() {
    int referenceString[30], pageFaults = 0, m, n, s, pages, frames;
    int flag = 0;
    printf("Enter number of frames: ");
    scanf("%d", &frames);
    printf("Enter number of pages: ");
    scanf("%d", &pages);
    printf("Enter reference string values: ");
    for(m = 0; m < pages; m++) {
        scanf("%d", &referenceString[m]);
    }
    int *temp = (int *)malloc(frames * sizeof(int));
    for(m = 0; m < frames; m++) {
        temp[m] = -1;
    }
    printf("\n\\n=====
\\n");
    for(int i = 0; i < frames; i++) {
        printf("frame[%d]\\t", i );
    }
    printf("page fault");
    printf("\n=====\\n");
    for(m = 0; m < pages; m++) {

```

```

    s = 0;
    for(n = 0; n < frames; n++) {
        if(referenceString[m] == temp[n]) {
            s++;
            pageFaults--;
            flag = 1;
        }
    }
    pageFaults++;
    if((pageFaults <= frames) && (s == 0)) {
        temp[m] = referenceString[m];
    } else if(s == 0) {
        temp[(pageFaults - 1) % frames] = referenceString[m];
    }
    printf("\n");
    for(n = 0; n < frames; n++) {
        printf("%4d\t\t", temp[n]);
    }
    if (flag == 0) {
        printf(" F" );
    }
    flag = 0;
}
printf("\n=====\\n");
printf("\nTotal Page Faults: %4d\\n", pageFaults);
free(temp);
return 0;
}

```

```

Enter number of frames: 4
Enter number of pages: 10
Enter reference string values: 0 1 7 2 3 2 7 1 0 3

=====
frame[0]    frame[1]    frame[2]    frame[3]    page fault
=====
0           -1         -1         -1         F
0           1         -1         -1         F
0           1         7         -1         F
0           1         7         2         F
3           1         7         2         F
3           1         7         2
3           1         7         2
3           1         7         2
3           0         7         2         F
3           0         7         2
=====

Total Page Faults:    6
PS C:\Users\samir\Desktop\project>

```

3. Write a program in C to implement LRU page replacement algorithm and calculate page faults.

```

#include <stdio.h>

int findLRU(int time[], int n) {
    int i, minimum = time[0], pos = 0;
    for(i = 1; i < n; ++i) {
        if(time[i] < minimum) {
            minimum = time[i];
            pos = i;
        }
    }
    return pos;
}

int main() {
    int no_of_frames, no_of_pages, frames[10], ref_string[30], counter = 0,
    time[10];
    int flag1, flag2;
    int i, j, pos, faults = 0;
    int flag = 0; // for hit if flag = 1 ; then hit else fault

    printf("Enter number of frames: ");
    scanf("%d", &no_of_frames);
    printf("Enter number of pages: ");
    scanf("%d", &no_of_pages);
    printf("Enter reference string: ");
    for(i = 0; i < no_of_pages; ++i) {
        scanf("%d", &ref_string[i]);
    }
    for(i = 0; i < no_of_frames; ++i) {
        frames[i] = -1;
    }

    printf("=====\n");
    for(i = 0; i < no_of_frames; i++) {
        printf("frame[%d]\t", i);
    }
    printf("page fault");
    printf("\n=====
\n");

    for(i = 0; i < no_of_pages; ++i) {
        flag1 = flag2 = 0;
        for(j = 0; j < no_of_frames; ++j) {
            if(frames[j] == ref_string[i]) {
                counter++;
                time[j] = counter;
                flag1 = flag2 = 1;
            }
        }
    }
}

```

```

        flag = 1; // hit case
        break;
    }
}
if(flag1 == 0) {
    for(j = 0; j < no_of_frames; ++j) {
        if(frames[j] == -1) {
            counter++;
            faults++;
            frames[j] = ref_string[i];
            time[j] = counter;
            flag2 = 1;
            break;
        }
    }
}
if(flag2 == 0) {
    pos = findLRU(time, no_of_frames);
    counter++;
    faults++;
    frames[pos] = ref_string[i];
    time[pos] = counter;
}
printf("\n");
for(j = 0; j < no_of_frames; ++j) {
    printf("%4d\t\t", frames[j]);
}
if(flag == 0) {
    printf(" F");
}
flag = 0;
}
printf("\n=====
\n");
printf("\n\nTotal Page Faults = %d\n", faults);
return 0;
}

```

```

Enter number of frames: 4
Enter number of pages: 10
Enter reference string: 0 1 7 2 3 2 7 1 0 3
=====
frame[0]      frame[1]      frame[2]      frame[3]      page fault
=====
0             -1           -1           -1           F
0             1            -1           -1           F
0             1            7            -1           F
0             1            7            2           F
3             1            7            2           F
3             1            7            2
3             1            7            2
3             1            7            2
0             1            7            2           F
0             1            7            3           F
=====
Total Page Faults = 7
PS C:\Users\samir\Desktop\project>

```

4. Write a program in C to implement Clock Page replacement algorithm and calculate page faults.

```

#include <stdio.h>
#include <stdlib.h>

int main() {
    int frameCount, referenceCount, faults = 0, i, j, k = 0;
    printf("Enter number of frames: ");
    scanf("%d", &frameCount);
    int *frame = (int *)malloc(frameCount * sizeof(int));
    int *usebit = (int *)malloc(frameCount * sizeof(int));
    for (i = 0; i < frameCount; i++) {
        frame[i] = -1;
        usebit[i] = 0;
    }
    printf("Enter number of pages: ");
    scanf("%d", &referenceCount);
    int *reference = (int *)malloc(referenceCount * sizeof(int));
    printf("Enter reference string: ");
    for (i = 0; i < referenceCount; i++) {
        scanf("%d", &reference[i]);
    }
    printf("\n");
    printf("=====\n");
    printf("Clock Page Replacement Algorithm Output\n");
    printf("=====\n");
    printf("Frame\tPage\t\tPage Fault\n");
    printf("-----\t----\t\t-----\n");
    for (i = 0; i < referenceCount; i++) {
        int found = 0;
        for (j = 0; j < frameCount; j++) {
            if (frame[j] == reference[i]) {
                found = 1;
                usebit[j] = 1;
            }
        }
        if (!found) {
            faults++;
        }
    }
    printf("Total Page Faults = %d", faults);
}

```

```

        break;
    }
}
if (!found) {
    while (usebit[k] == 1) {
        usebit[k] = 0;
        k = (k + 1) % frameCount;
    }
    frame[k] = reference[i];
    usebit[k] = 1;
    k = (k + 1) % frameCount;
    faults++;
}
printf("%d\t", i + 1);
for (j = 0; j < frameCount; j++) {
    if (frame[j] == -1) {
        printf(" \t");
    } else {
        printf("%d\t", frame[j]);
    }
}
if (!found) {
    printf("\tF");
}
printf("\n");
}
printf("-----\t----\t\t-----\n");
printf("\nTotal Page Faults: %d\n", faults);
free(frame);
free(usebit);
free(reference);
return 0;
}

```

```

PS C:\Users\samir\Desktop\project> cd "c:\Users\samir\Desktop\project\" ; if ($?) { gcc lab6.c -o lab6 } ; if ($?) { .\lab6 }
Enter number of frames: 4
Enter number of pages: 10
Enter reference string: 0 1 7 2 3 2 7 1 0 3

=====
Clock Page Replacement Algorithm Output
=====

```

Frame	Page	Page Fault
1	0	F
2	0 1	F
3	0 1 7	F
4	0 1 7 2	F
5	3 1 7 2	F
6	3 1 7 2	
7	3 1 7 2	
8	3 1 7 2	
9	3 0 7 2	F
10	3 0 7 2	

```

-----
Total Page Faults: 6
PS C:\Users\samir\Desktop\project>

```